

2D Criteria and Pseudo-3D Criteria	Data link	Formats	Interpolation	Interpolation resolution	Normalization	MCDA
Bouguer Anomaly	https://doi.org/10.15121/1452733	Shapefile (.shp) - Point	Neighbors	10 m	Linear	Gamma Fuzzy
Distance to Faults	https://doi.org/10.15121/1452761	Shapefile (.shp) - Line	Euclidian Distance	10 m		
Distance to Epicenters	https://opendata.gis.utah.gov/datasets/af0cbb8ecca942d58367f68bcd2643b6_0/explore?location=39.368893%2C-110.848500%2C-1.00	Shapefile (.shp) - Point	Euclidian Distance	10 m		
Heat Flow	https://doi.org/10.15121/1405032	Shapefile (.shp) - Point	Neighbors	10 m		
Land Surface Temperature	https://github.com/marcuslucamaral/Utah_FORGE_Geothermal_MCDA_Fuzzy/tree/main/LST	Raster	Raster Resolution: 30 m	10 m		
P-Wave Velocity	https://doi.org/10.15121/1989942	ASCII text (.dat)	Neighbors	10 m		
Resistivity	https://doi.org/10.15121/2283064	ASCII text (.dat)	Neighbors	10 m		
Density	https://doi.org/10.15121/1542061	ASCII text, comma-separated (.csv)	Neighbors	10 m	Discrete/ binary normalization (Basian = 0.2; Granitoid = 0.8)	
Basement Boundary	https://doi.org/10.15121/1495398	ASCII text, comma-separated (.csv)	None (binary rasterization)	10 m		

All GIS procedures were conducted using **ArcGIS (ArcMap module)**, with the following tools and operations:

• **Interpolation Tools-**

– *Natural Neighbor (Spatial Analyst) → Output cell size: 10*

For more details about the tool: <https://pro.arcgis.com/en/pro-app/latest/tool-reference/spatial-analyst/natural-neighbor.htm>

– *Euclidean Distance (Spatial Analyst) → Output cell size: 10*

For more details about the tool: <https://pro.arcgis.com/en/pro-app/latest/tool-reference/spatial-analyst/euclidean-distance.htm>

• **Normalization Tools**

– *Fuzzy Membership (Spatial Analyst) → Membership type: Linear*

For more details about the tool: <https://pro.arcgis.com/en/pro-app/latest/tool-reference/spatial-analyst/fuzzy-membership.htm>

- **MCDA**

- Fuzzy Overlay (Spatial Analyst) → *Overlay type: Gamma*

For more details about the tool: <https://pro.arcgis.com/en/pro-app/latest/tool-reference/spatial-analyst/fuzzy-overlay.htm>